MACHINE LEARNING APPROACHES FOR DEPRESSION DETECTION USING EEG DATA

Running head: Machine learning and depression

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ABSTRACT

Background: Major depressive disorder is a major contributor to the global disease burden. Traditional assessment methods rely on clinical evaluation, medical history, and self-reported measures, which are subjective, time-consuming, and difficult to scale. Electroencephalography (EEG), a non-invasive method for measuring brain activity, offers potential for objective depression detection, though its signal processing is complex. Machine learning techniques have been increasingly applied to EEG data, but studies vary widely in algorithms, electrode configurations, and evaluation metrics. This study aims to identify the most commonly used machine learning algorithms, typical EEG electrode set-ups in resting-state recordings, the most frequently reported performance metrics, and the overall effectiveness of these approaches.

Methods: A literature search was conducted in PubMed to identify original studies published within the past five years, using the Medical Subject Headings (MeSH) terms "Electroencephalography," "Major Depressive Disorder," and "Machine Learning." The review followed the PIRD framework and included studies comparing individuals with depression to healthy controls. Both supervised machine learning and deep learning approaches were considered eligible; however, only findings from supervised machine learning methods were analyzed and summarized. Studies based solely on task- or stimulus-evoked EEG data, such as event-related potentials (ERPs), were excluded.

Results: Twenty studies were included. Support vector machines were the most commonly used algorithm, followed by random forests, with several other methods applied less frequently. Accuracy was the most reported performance metric, along with sensitivity and specificity, while fewer studies reported F1-score or AUC. The number of EEG electrodes varied widely across studies. Overall, machine learning models showed high performance, though results varied considerably, with a mean accuracy of 83.8%.

Conclusion: In conclusion, current evidence suggests that EEG-based machine learning approaches hold significant promise for depression detection. However, the literature is marked by substantial heterogeneity in the algorithms applied, the performance metrics reported, and EEG acquisition protocols, particularly in the wide variation in the number of electrodes used for signal acquisition.

INTRODUCTION

Major depressive disorder is a clinical syndrome characterized by the presence of at least 5 of 9 diagnostic symptoms persisting for a minimum of 2 weeks, including either depressed mood or anhedonia. It is a leading condition to the global burden of disease, affecting individuals across the lifespan, both sexes, and diverse geographic regions worldwide [1]. In 2019, the Global Burden of Diseases, Injuries, and Risk Factors Study identified major depressive disorder as the leading cause of disability among mental disorders, ranking it among the top contributors to global disease burden [2].

Current approaches to the assessment of depressive disorders rely on clinical evaluation of patients' mental status, medical history, and self-reported instruments, such as the Patient Health Questionnaire–9 (PHQ-9) [3,4]. These methods are inherently subjective, time-intensive, and not readily scalable for repeated assessments. Consequently, contemporary psychiatric evaluation may be limited in its ability to provide consistent, precise, and individualized measurement of depressive symptoms. In addition, the global lack of mental health professionals represents another crucial barrier to the early detection of depression [5,6]. These limitations underscore the need for scalable, objective adjunctive methods to facilitate the early detection of depression, enabling timely intervention and reducing the risk of adverse clinical outcomes and long-term complications.

Electroencephalography (EEG) is a non-invasive technique used to measure electrical activity of the brain. It involves placing electrodes on the scalp to detect the brain's electrical impulses, which reflect neuronal activity [7]. The processing of EEG signals has become a promising tool in the diagnosis of neurological and psychiatric disorders, offering objective, quantifiable biomarkers of brain function that may enhance early detection, improve diagnostic accuracy, and support more personalized treatment

approaches [8-11]. However, EEG signal processing is complex, and advanced computational approaches, including machine learning techniques, have been increasingly applied to improve the detection of depression.

Machine learning refers to a class of computational methods that enable algorithms to identify patterns in data and generate predictions without being explicitly programmed [12]. In the context of depression, numerous studies have applied machine learning techniques to EEG data to support detection and classification [10,13-16]. However, these studies have used heterogeneous approaches, including different machine learning algorithms, varying numbers and configurations of EEG electrodes, and diverse performance metrics. This methodological variability limits comparability across studies and makes difficult efforts to standardize and establish the overall efficacy of EEG-based machine learning approaches for depression detection. In this sense, the present study aimed to identify (1) the most frequently used machine learning algorithms, (2) the number of EEG electrodes employed for signal acquisition at resting state, (3) the performance metrics most reported, and (4) the overall performance of these machine learning approaches.

METHODS

A literature search was conducted in PubMed to identify original studies published in the last five years. The search strategy used the Medical Subject Headings (MeSH) terms: (("Electroencephalography"[Mesh]) AND "Major Depressive Disorder"[Mesh]) AND "Machine Learning"[Mesh]. The review was structured following the PIRD framework, focusing on Population (individuals diagnosed with major depressive disorder), Index test (resting-state EEG–based machine learning approaches), Reference standard (comparison between depression and non-depression individuals), and

Diagnostic outcomes (accuracy, sensitivity, specificity, and other performance metrics). Studies were included if they compared participants with depression to healthy control groups. Both supervised machine learning and deep learning algorithms were considered eligible; however, only the results of supervised machine learning algorithms are described and summarized in this review. Studies that used only task- or stimulus-evoked EEG, such as event-related potentials (ERPs), were excluded.

Sensitivity was considered the primary performance metric, as it reflects the ability of machine learning algorithms to correctly identify individuals with depression. This focus aligns with the clinical objective of using the machine learning approach as an auxiliary diagnostic tool, where early and accurate detection of depressive symptoms is critical for timely intervention and for reducing the risk of progression and related complications. By prioritizing sensitivity, the evaluation emphasizes minimizing false negatives, which is particularly important in a mental health context where missed cases can lead to delayed treatment and adverse outcomes.

All screening, selection, and extraction of information from the included studies were conducted using Rayyan (Rayyan Systems Inc.), ensuring a systematic and reproducible review process.

RESULTS

Following the systematic search, a total of 65 articles were initially identified. After removing duplicates, the remaining records underwent screening based on titles and abstracts, resulting in 25 potentially relevant studies. These studies were then assessed in full text for eligibility. After full-text review, 20 studies met all inclusion criteria and were included in the final analysis for data extraction. This stepwise screening process ensured

that only studies directly relevant to resting-state EEG–based machine learning approaches for depression detection were analyzed. Figure 1 displays the detailed flowchart screening process.

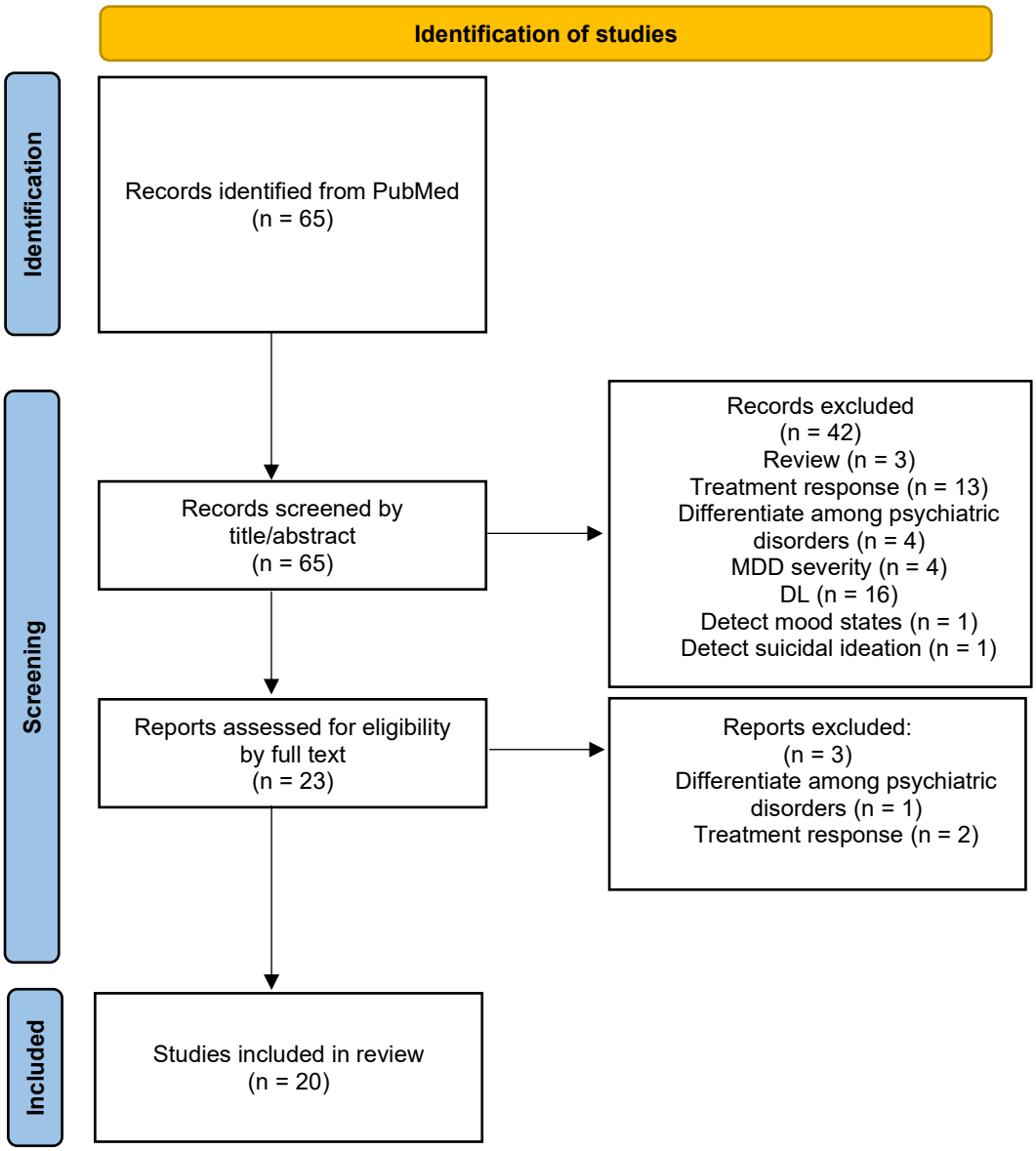

Figure 1. Flowchart of study identification, screening, eligibility, and inclusion.

The geographic distribution of the studies included in the analysis reveals a clear concentration in a limited number of countries, with the highest frequency observed in the United States. Moderate representation is seen in parts of Western Europe, particularly

in the United Kingdom, the Netherlands, Switzerland, and Ireland. Outside of these regions, there is little to no representation, indicating a pronounced geographic imbalance in research available in the PubMed related to EEG and machine learning for depression detection over the past five year (Figure 2).

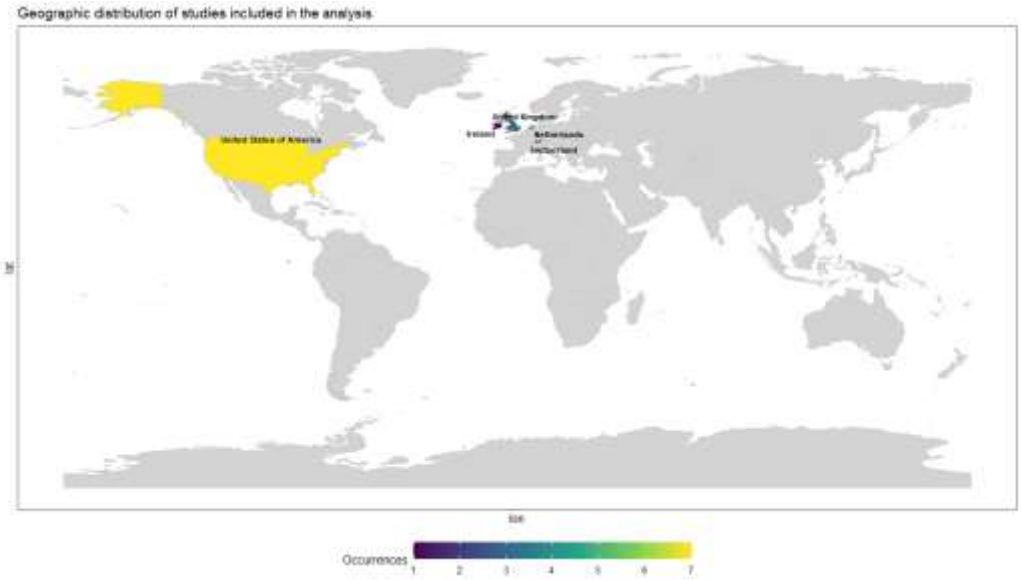

Figure 2. Geographic distribution of studies included in the analysis.

Support vector machine was the most frequently used algorithm, identified in 15 studies, followed by random forest in 9 studies. k-Nearest neighbors and logistic regression were each used in 5 studies. Naive Bayes was reported in 4 studies, while gradient boosting and linear discriminant analysis were each reported in 3 studies. Decision tree and XGBoost were each used in 2 studies, whereas RUSBoost, quadratic discriminant analysis, and GentleBoost were each identified in 1 study. The distribution of machine learning algorithms across the included studies is shown in Figure 3.

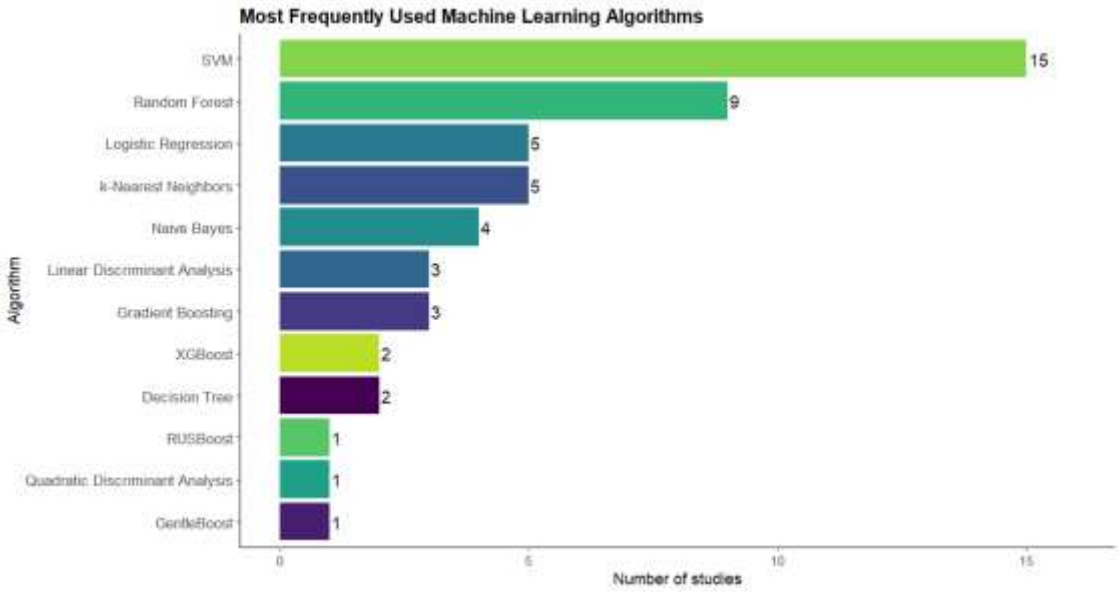

Figure 3. Frequency of machine learning algorithms used across the included studies.

Regarding performance metrics reported across studies, accuracy was the most reported performance metric across the included studies, appearing in 18 studies. Sensitivity and specificity were reported in 13 and 12 studies, respectively. F1-score was reported in 9 studies, whereas receiver operating characteristic area under the curve was reported in 4 studies. Figure 4 shows the distribution of performance metrics reported across the included studies.

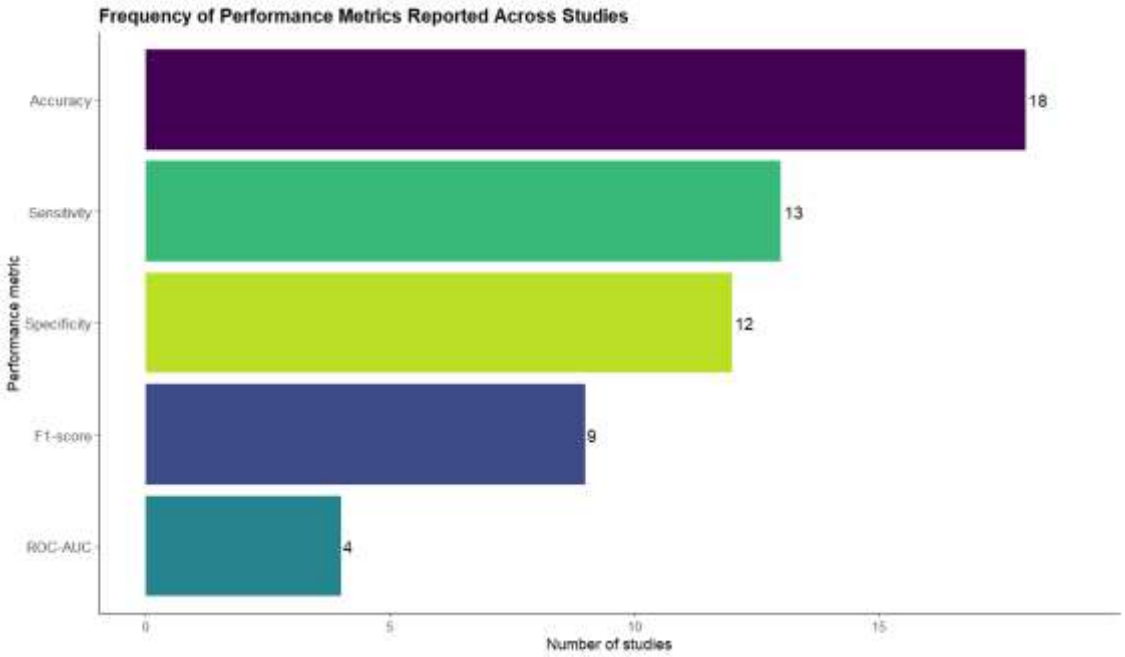

Figure 4. Frequency of performance metrics used across the included studies.

The number of EEG electrodes used for signal acquisition varied widely across the included studies, ranging from 2 to 64 electrodes. Nineteen electrodes were the most used configuration, appearing in 9 studies. Sixty-four electrodes were reported in 3 studies, while 32 and 62 electrodes were each reported in 2 studies. Other configurations, including 2, 20, 21, and 42 electrodes, were each used just in 1 study. These findings underscore the substantial heterogeneity in EEG acquisition protocols across studies. Figure 5 shows the distribution of performance metrics reported across the included studies

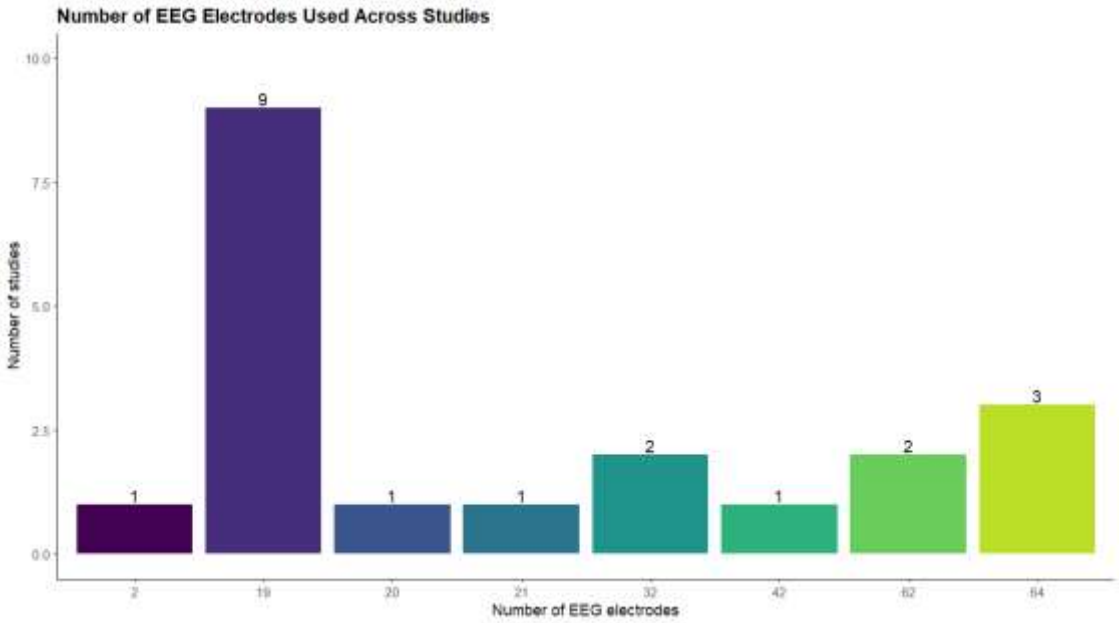

Figure 5. Number of EEG electrodes used across the included studies.

Across the included studies, reported performance of the machine learning approaches was generally high, although substantial variability was observed across models and studies. Considering all reported algorithm-specific performance values in the extracted dataset, the mean performance was 83.8%, with values ranging from 48.0% to 99.0%. Among the most frequently evaluated approaches, support vector machine showed a mean performance of 88.2% (range, 67.0%-99.0%), random forest 79.0% (range, 60.0%-98.0%), logistic regression 78.2% (range, 68.0%-86.0%), and k-nearest neighbors 69.8% (range, 48.0%-88.0%). These findings suggest overall good discriminative performance, but with marked heterogeneity across machine learning approaches and study settings. Table 1 summarizes the machine learning algorithms, performance metrics, number of EEG electrodes used, and reported performance across the included studies.

Table 1. Summary of machine learning algorithms, performance metrics, and reported performance across the included studies

First author	Year	Algorithms used	Metrics used	Number of EEG electrodes	Performance for each algorithm (Sensibility)
Aderinwale et. al. [13]	2023	Support vector machine	Accuracy*	2	Support vector machine: 90%
Margarette Sanchez et. al. [10]	2022	Support vector machine; Random forest; k-Nearest neighbors	Accuracy; Sensitivity; Specificity	42	Support vector machine: 83%; Random forest: 83%; k-Nearest neighbors: 83%
Earl et. al. [17]	2024	Random forest	Accuracy; F1-score; ROC-AUC	19	Random forest: 60%
Shim et. al. [18]	2023	Support vector machine	Accuracy; Sensitivity; Specificity	64	Support vector machine: 83%
Saeedi et. al. [19]	2020	Support vector machine; k-Nearest neighbors; Enhanced k-nearest neighbors; Multilayer perceptron[#]	Accuracy; Sensitivity; Specificity	19	Support vector machine: 96%; k-Nearest neighbors: 88%; Enhanced k-nearest neighbors: 97%
McVoy et. al. [20]	2025	Random forest	Sensitivity; Specificity; ROC-AUC	32	Random forest: 65%
Chen et. al. [21]	2024	Random forest	Accuracy; Sensitivity; Specificity; F1-score	21	Random forest: 75%

Movahed et. al. [22]	2021	Support vector machine; Linear SVM; RBF SVM; Random forest; Decision tree; Naive Bayes; Logistic regression; RUSBoost; GentleBoost	Accuracy; Sensitivity; Specificity; F1-score	20	Support vector machine: 99%; Linear SVM: 98%; RBF SVM: 99%; Random forest: 98%; Decision tree: 92%; Naive Bayes: 93%; Logistic regression: 86%; RUSBoost: 93%; GentleBoost: 93%
Khadidos et. al. [23]	2023	Random forest; Gradient boosting; Long short-term memory [#]; Convolutional neural network [#]; Gated recurrent unit [#]; XGBoost	Accuracy; Sensitivity; F1-score; ROC-AUC	19	Random forest: 96%; Gradient boosting: 92%; XGBoost: 97%
Jang et. al. [24]	2023	Support vector machine; Linear discriminant analysis	Accuracy; Sensitivity; Specificity	62	Support vector machine: 87%; Linear discriminant analysis: 84%
Li et. al. [25]	2022	Support vector machine; Logistic regression	Accuracy; Sensitivity; Specificity; F1-score	19	Support vector machine: 97%; Logistic regression: 82%
Zandbagleh et. al. [14]	2024	Support vector machine; Logistic regression	ROC-AUC	64	Support vector machine: 86%; Logistic regression: 82%
Wu et. al. [26]	2021	Support vector machine; k-Nearest neighbors; Linear discriminant analysis	Accuracy; Sensitivity; Specificity	32	Support vector machine: 88%; k-Nearest neighbors: 48%; Linear discriminant analysis: 75%
Hong et. al. [27]	2021	Support vector machine; Random forest; Logistic regression	Accuracy*; F1-score	19	Support vector machine: 90%; Random forest: 73%; Logistic regression: 73%

Song et. al. [15]	2025	Quadratic discriminant analysis	Accuracy; Sensitivity; Specificity; F1-score	62	Quadratic discriminant analysis: 70%
Mulc et. al. [28]	2025	Support vector machine; Random forest; k-Nearest neighbors; Decision tree; Naive Bayes; XGBoost	Accuracy*; F1-score	19	Support vector machine: 78% Random forest: 73%; k-Nearest neighbors: 60%; Decision tree: 78%; Naive Bayes: 75%; XGBoost: 80%
Sarisik et. al. [29]	2025	Support vector machine	Accuracy; Sensitivity; Specificity	19	Support vector machine: 67%
Baghernezhad et. al. [16]	2025	Support vector machine	Accuracy*	19	Support vector machine: 93%
Liu et. al. [30]	2022	Linear SVM; RBF SVM; Linear discriminant analysis; Logistic regression	Accuracy; Sensitivity; Specificity	64	Linear SVM: 98%; RBF SVM: 82%; Linear discriminant analysis: 78%; Logistic regression: 68%
He et. al. [31]	2025	Support vector machine; Random Forest; k-Nearest neighbors; Multilayer perceptron [#]; Naive Bayes; Gradient boosting	Accuracy*; F1-score	19	Support vector machine: 88%; Random forest: 88%; k-Nearest neighbors: 81%; Naive Bayes; 57% Gradient boosting: 85%

* Indicates the metric used when sensitivity was not reported; # Deep learning method;

DISCUSSION

The current study aimed to identify (1) the most frequently used machine learning algorithms, (2) the number of EEG electrodes employed for signal acquisition, (3) the performance metrics most reported, and (4) the overall performance of these machine learning approaches to detect depression. We found that support vector machine was the most frequently used algorithm across studies, that the number of EEG electrodes used for signal acquisition varied widely from 2 to 64, and that accuracy was the most frequently reported performance metric. Overall, machine learning approaches applied to EEG data showed generally good performance for depression detection, although substantial heterogeneity was observed across algorithms, performance metrics, and EEG acquisition protocols. This variability may have contributed to between-study differences in model performance and limits direct comparability across studies.

The predominance of support vector machine across the included studies may be partly explained by its suitability for high-dimensional classification problems, particularly when the number of extracted EEG features is large relative to the sample size [32]. In the context of depression detection using resting-state EEG, this characteristic is especially relevant because studies often rely on multiple signal-derived measures obtained from a limited number of participants [10,13,14,16,18,19,22,24-29,31]. In general, support vector machine is a margin-based machine learning method designed to identify an optimal separating boundary between classes and can perform well even when the underlying decision boundary is not strictly linear, depending on the kernel used [12,33], which may be advantageous for depression detection using resting-state EEG.

Other frequently reported algorithms, such as k-nearest neighbors and logistic regression, reflect different trade-offs between flexibility, simplicity, and interpretability. k-Nearest neighbors is a distance-based approach that can be appealing because of its

simplicity [12]; however, its performance may be more sensitive to feature scaling, noise, and the structure of the input space, which may limit stability in heterogeneous EEG datasets [34]. Logistic regression, in contrast, is a more parsimonious and interpretable model [12], and its inclusion in several studies may reflect its usefulness as a baseline classifier against more complex machine learning methods. Similarly, linear and quadratic discriminant methods offer relatively straightforward classification frameworks but may be more dependent on assumptions regarding data distribution and class structure. Furthermore, the performance of these machine learning algorithms was generally lower than that observed for support vector machine.

Tree-based boosting approaches and probabilistic classifiers, including gradient boosting, XGBoost, Naive Bayes, RUSBoost, and GentleBoost, were less frequently reported but further illustrate the methodological diversity of the field. These methods may offer advantages in specific analytical settings, such as improved handling of nonlinearities, class imbalance, or computational efficiency [35-38]. However, the marked variation in their reported performance suggests that algorithm choice alone is unlikely to explain differences across studies. Instead, model performance probably also reflects broader methodological factors, including sample composition, diagnostic criteria, EEG preprocessing pipelines, feature engineering strategies, electrode configuration, validation procedures, and the metric selected to summarize classification performance

This review has important limitations. The search strategy was restricted to PubMed, which may have resulted in the omission of relevant studies indexed in other databases. In addition, substantial heterogeneity across studies in EEG acquisition protocols, electrode configuration, preprocessing pipelines, feature extraction methods, machine learning algorithms, validation strategies, and reported performance metrics limited direct comparisons between studies. The generally small sample sizes of several

included studies may also have reduced the robustness and generalizability of the reported results. Accordingly, these findings should be interpreted with caution.

In conclusion, the current evidence indicates that EEG-based machine learning approaches have considerable potential for depression detection, with generally good, reported performance across studies. However, the literature is characterized by substantial heterogeneity in the algorithms used, the performance metrics reported, and EEG acquisition protocols, including marked variability in the number of electrodes employed for signal acquisition. This methodological diversity limits direct comparison across studies and may hinder the translation of these approaches into clinical settings. Taken together, these findings suggest that EEG-based machine learning approaches are promising tools for depression detection, but also highlight the need for greater methodological consistency to improve interpretability, reproducibility, and clinical translation.

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